Gödel–Dummett–Gödel–Löb Logic

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Gödel–Dummet Logic

- Intermediate logics satisfying

$$(\varphi \to \psi) \vee (\psi \to \varphi).$$

- Introduced by Gödel to prove intuitionistic logic is not finite valued.
- Later axiomatized by Dummett.
- Semantics include both relational and real-valued Kripke frames.

Gödel–Dummet Modal Logic

- First studied by Caicedo and Rodriguez.
- Closely related to intuitionistic modal logics.
- Semantics include both bi-relational and real-valued Kripke frames.

Gödel–Dummett meets Gödel–Löb

Why study a Gödel–Dummett–Gödel–Löb Logic?

- The logic's name.
- Closely related to IGL.

Our results:

- Equivalence between bi-relational and real-valued semantics.
- Completeness w.r.t. non-well-founded and cyclic *fully labeled* proof systems.
- Finite model and decidability.

fmp and decidability of IGL also follows from our methods.

Real-valued Kripke models

A GK-model is a triple $\langle W, R, e \rangle$ where:

- W is the set of possible worlds;
- $R : W \times W \to [0, 1]$ is the accessibility function;
- $e : W \times \mathrm{Prop} \to [0, 1]$ is the valuation function.

Valuation

Let

$$x \Rightarrow y := \begin{cases} 1 & \text{if } x \leq y; \\ y & \text{otherwise} \end{cases}$$

Then extend e to all formulas by:

- $e(w, \bot) := 0$;
- $e(w, \varphi \wedge \psi) := \min\{e(w, \varphi), e(w, \psi)\}$;
- $e(w, \varphi \vee \psi) := \max\{e(w, \varphi), e(w, \psi)\}$;
- $e(w, \varphi \to \psi) := e(w, \varphi) \Rightarrow e(w, \psi)$;
- $e(w, \Box\varphi) := \inf_{v \in W}\{R(w, v) \Rightarrow e(v, \varphi)\}$; and
- $e(w, \Diamond\varphi) := \sup_{v \in W}\{\min\{R(w, v), e(v, \varphi)\}\}$.

Real-valued GGL-models

A GGL-model is a GK-model $\langle W, R, e\rangle$ where:

- $\min\{R(w,v), R(v,u)\} \leq R(w,v)$;
- there is no sequence $w_0, w_1, w_2, \ldots$ such that

$$R(w_i, w_{i+1}) > 0 \text{ for all } i \in \omega.$$

Proposition

$\Box(\Box P \to P) \to \Box P$ is valid over all GGL-models; that is,

$$e(w, \Box(\Box P \to P) \to \Box P) = 1$$

for all GGL-model M and world w.

Converse well-foundedness requirement

Let $M = \langle W, R, e \rangle$ be defined by:

- $W = \{w_0, w_1, w_2, \dots\}$;
- $w_i R w_j = \frac{1}{j}$, if $i < j$; $w_i R w_j = 0$, otherwise;
- $w(w_i, P) = \frac{1}{i+1}$.

Then

$$e(w_i, \Box P) = \inf_{j>i}\{w_i R w_j \Rightarrow e(w_i, P)\}$$
$$= \inf_{j>i}\left\{\frac{1}{i} \Rightarrow \frac{1}{i+1}\right\} = \inf_{j>i}\{\frac{1}{i+1}\} = 0.$$

So $e(w_i, \Box P \to P) = 0 \Rightarrow \frac{1}{i+1} = 0$ and

$$e(w_0, \Box(\Box P \to P)) = \inf_{i>0}\{w_0 R w_i \Rightarrow e(w_i, \Box P \to P)\}$$
$$= \inf_{i>0}\left\{\frac{1}{i} \Rightarrow 1\right\} = 1.$$

So $e(w_0, \Box(\Box P \to P) \to \Box P)$ is 0.

Bi-relational Kripke models

A GGL-model is a triple $\langle W, \preceq, R, V \rangle$ where:

- W is the set of possible worlds;
- $R \subseteq W \times W$ is the accessibility function;
- $V : \mathrm{Prop} \to \mathcal{P}(W)$ is the valuation function;
- $V(P)$ is upwards closed w.r.t. $\preceq$;
- R is forward and backward confluent;
- $\preceq$ is locally linear.

Valuation

Let $M = \langle W, \preceq, R, V \rangle$ be a bi-relational GGL-model.

- $M, w \models P$ iff $w \in V(P)$;
- $M, w \models \bot$ never holds;
- $M, w \models \varphi \wedge \psi$ iff $M, w \models \varphi$ and $M, w \models \psi$;
- $M, w \models \varphi \vee \psi$ iff $M, w \models \varphi$ or $M, w \models \psi$;
- $M, w \models \varphi \to \psi$ iff, for all $v \in W$, if $w \preceq v$ and $M, v \models \varphi$, then $M, v \models \psi$;
- $M, w \models \Box\varphi$ iff, for all $v, u \in W$, if $w \preceq v$ and vRu, then $M, u \models \varphi$; and
- $M, w \models \Diamond\varphi$ iff there is v such that wRv and $M, v \models \varphi$.

Bi-relational models to real-valued models

- Let $M = \langle W, \preceq, R, V \rangle$ be a finite *layered* bi-relational GGL-model.
- We define $M^{\mathbb{R}} = \langle W^{\mathbb{R}}, R^{\mathbb{R}}, V^{\mathbb{R}} \rangle$ as follows:
  - $W^{\mathbb{R}}$ is the connected classes $[w]$ w.r.t. $\preceq$;
  - $R([w],[v]) := \frac{\#(\text{layers where a member of } [w] \text{ accesses a member of } [v])}{\#(\text{total of layers})}$;
  - $e([w],P) := \frac{\#(\text{layers where a member of } [w] \text{ satisfies } P)}{\#(\text{total of layers})}$.
- If a formula is not valid over bi-relational models, we can find a real-valued model refuting it.

Real-valued models to bi-relational models

- Let $M = \langle W, R, e \rangle$ be a real-valued GGL-model.
- We define $M^{\mathbb{B}} = \langle W^{\mathbb{B}}, \preceq, R^{\mathbb{B}}, V^{\mathbb{B}} \rangle$ as follows:
  - $W^{\mathbb{B}}$ is $\{\langle w, r \rangle \mid w \in W, r \in [0,1]\}$;
  - $\langle w, r \rangle \preceq \langle w', r' \rangle$ iff $w = w'$ and $r' < r$;
  - $\langle w, r \rangle R \langle w', r' \rangle$ iff $r = r'$ and $R(w, w') > r$;
  - $\langle w, r \rangle \in V(P)$ iff $e(w, P) > r$.
- $M^{\mathbb{B}}, \langle w, r \rangle \models \varphi$ iff $e(w, \varphi) > r$.
- If a formula is not valid over real-valued models, we can find a bi-relational model refuting it.

Labeled sequents

- Labeled formula:

$$x : \varphi,$$

  where x is some variable and φ a formula.
- Fully labeled sequent:

$$\mathbf{R}, \Gamma \vdash \Delta,$$

  where:
  - $\mathbf{R}$ is a finite collection of statements of the forms

$$x \preceq y \text{ and } xRy$$

    relating variables;
  - Γ and Δ are finite collections of labeled formulas.

Some inference rules — I

$$\text{id}\ \frac{}{\mathbf{R}, \Gamma, x : P \vdash \Delta, x : P}$$

$$\text{pres}\ \frac{\mathbf{R}, x \preceq y, x : \varphi, y : \varphi, \Gamma \vdash \Delta}{\mathbf{R}, x \preceq y, x : \varphi, \Gamma \vdash \Delta}$$

$$\text{tr}\ \frac{\mathbf{R}, xRy, yRz, xRz, \Gamma \vdash \Delta}{\mathbf{R}, xRy, yRz, \Gamma \vdash \Delta}$$

$$\wedge\text{l}\ \frac{\mathbf{R}, \Gamma, x : A \wedge B, x : A, x : B \vdash \Delta}{\mathbf{R}, \Gamma, x : A \wedge B \vdash \Delta}$$

$$\rightarrow\text{l}\ \frac{\mathbf{R}, \Gamma, x : A \rightarrow B \vdash \Delta, x : A \qquad \mathbf{R}, \Gamma, x : A \rightarrow B, x : B \vdash \Delta}{\mathbf{R}, \Gamma, x : A \rightarrow B \vdash \Delta}$$

Some inference rules — II

$$\Diamond\mathrm{r}\ \frac{\mathbf{R}, \Gamma \vdash \Delta, x : \Diamond A, \{y : A \mid xRy\}}{\mathbf{R}, \Gamma \vdash \Delta, x : \Diamond A}$$

$$\Box\mathrm{l}\ \frac{\mathbf{R}, \Gamma, x : \Box A, \{y : A \mid xRy\} \vdash \Delta}{\mathbf{R}, \Gamma, x : \Box A \vdash \Delta}$$

$$\Diamond\mathrm{l}\ \frac{\mathbf{R}, xRy, \Gamma, x : \Diamond A, y : A \vdash \Delta}{\mathbf{R}, \Gamma, x : \Diamond A \vdash \Delta}\ (y \text{ is fresh})$$

Some inference rules — III

$$\mathsf{lin}\ \frac{\mathbf{R}, x \preceq y, x \preceq z, y \preceq z\Gamma, \vdash \Delta \qquad \mathbf{R}, x \preceq y, x \preceq z, z \preceq y\Gamma, \vdash \Delta}{\mathbf{R}, x \preceq y, x \preceq z, \Gamma, \vdash \Delta}$$

$$F_1\ \frac{\mathbf{R}, xRy, y \preceq z, x \preceq u, uRz, \Gamma, x : A \vdash x : B}{\mathbf{R}, \Gamma \vdash \Delta, x : A \to B}\ (u \text{ is fresh})$$

$$F_2\ \frac{\mathbf{R}, xRy, x \preceq z, y \preceq u, zRu, \Gamma, x : A \vdash x : B}{\mathbf{R}, \Gamma \vdash \Delta, x : A \to B}\ (u \text{ is fresh})$$

Some inference rules — IV

Rules which instantiate new $\prec$-atoms:

$$\rightarrow\mathrm{r}\ \frac{\mathbf{R}, x \preceq y, \Gamma, y : A \vdash \Delta, y : B, x : A \rightarrow B}{\mathbf{R}, \Gamma \vdash \Delta, x : A \rightarrow B}\ (y \text{ is fresh})$$

$$\Box\mathrm{r}\ \frac{\mathbf{R}, x \preceq y, yRz, \Gamma \vdash \Delta, z : A, x : \Box A}{\mathbf{R}, \Gamma \vdash \Delta, x : \Box A}\ (y, z \text{ is fresh})$$

Loops

A (progressing) loop v from a leaf $S = \mathbf{R}, \Gamma \vdash \Delta$ to a node below $S' + \mathbf{R}', \Gamma' \vdash \Delta'$ is a function such that:

- $v : \mathrm{Var}(S') \to \mathrm{Var}(S)$;
- if $x : \varphi \in \Gamma'$, then $v(x) : \varphi \in \Gamma$;
- if $x : \varphi \in \Delta'$, then $v(x) : \varphi \in \Delta$;
- if $xR'y$, then $v(x)Rv(y)$;
- if $x \preceq' y$, then $v(x) \preceq v(y)$;
- for each x in $\mathbf{R}'$, $v(x) \preceq x$ or $x \preceq; Rv(x)$
- there is $x \in \mathrm{Var}(R')$ such that $x \preceq; Rv(x) \in \mathbf{R}$.

cmℓGGL-proofs

A cyclic GGL-proofs are

- finitely branching tree respecting the inference rules;
- each leaf is an axiom or has a progressing loop to some node below.

We say GGL $\vdash \varphi$ if there is a cyclic GGL-proof whose root is $\vdash x : \varphi$.

Soundness

Theorem

Let T be a cmℓGGL*-proof of φ, then φ is valid over* GGL*-frames.*

Proof.

- Suppose otherwise, let $M, w \nVdash \varphi$.
- Note that each rule of the proof systems is sound.
- Going up level by level over T, we can build a sequence of interpretations σ_i taking the labels occurring in T to worlds in M.
- As T is a proof, the union $\bigcup \sigma_i$ witnesses an infinite $\preceq; R$-path in M.

$\square$

Proof/Countermodel Search

Fix a formula φ. We simultaneously search for either a countermodel or a cyclic proof for it.

- Start with a tree with root $x : \varphi$.
- While each of the leaves has a non-saturated layer, repeat operations to saturate the layer.
- If a layer gets too long, we have a loop. So we stop building the branch.
- Apply $\Box$r and $\to$r to generate new layers.
- If all the generated layers are redundant, we have a countermodel.
- If any layer is not redundant, keep building the tree.

In the end, we either get a cyclic proof or a countermodel.

Theorem

cmℓGGL *is complete w.r.t.* GGL*-models.*

As the countermodels built in the last slide are finite, we have:

Theorem

GGL *has the finite model property. Furthermore, we can suppose the finite model refuting some formula is layered.*

Theorem

GGL *is decidable.*

Results

Our results for GGL:

- Equivalence between bi-relational and real-valued semantics.
- Completeness w.r.t. non-well-founded and cyclic *fully labeled* proof systems.
- Finite model and decidability.

Again, we also get fmp and decidability for IGL.

Open Questions

Question

Is GK4 $+ \Box(\Box\varphi \to \varphi) \to \Box\varphi$ *complete w.r.t.* GGL*?*

I think no, but an axiomatization should be possible in a conservative extension.

Question

Do the proof systems have some form of interpolation?

Semantic methods are not viable (for now?). A proof via Maehara-method might work.

Question

Is GGL*'s decidability* PSPACE*-complete?*

No idea. Help welcome.